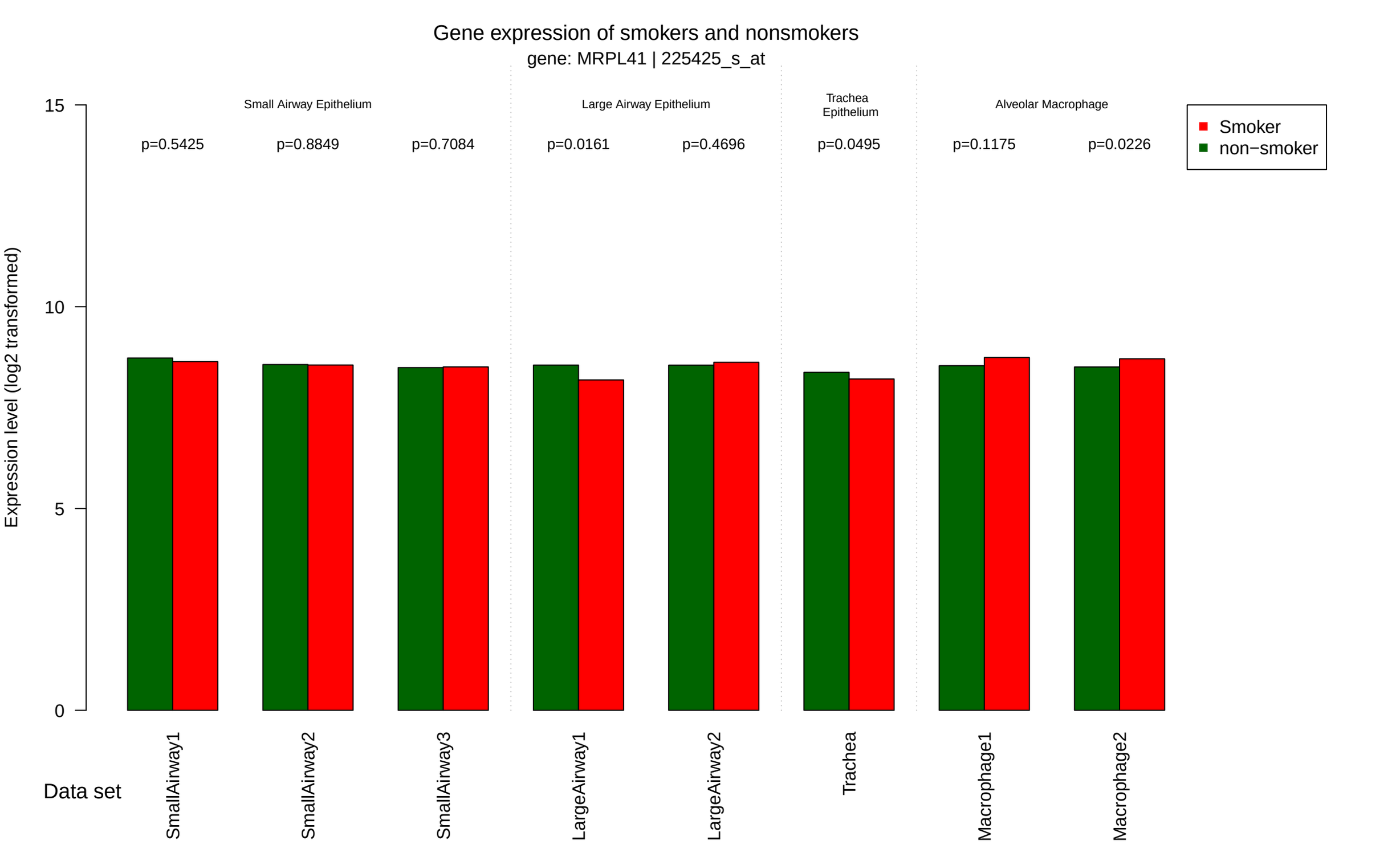
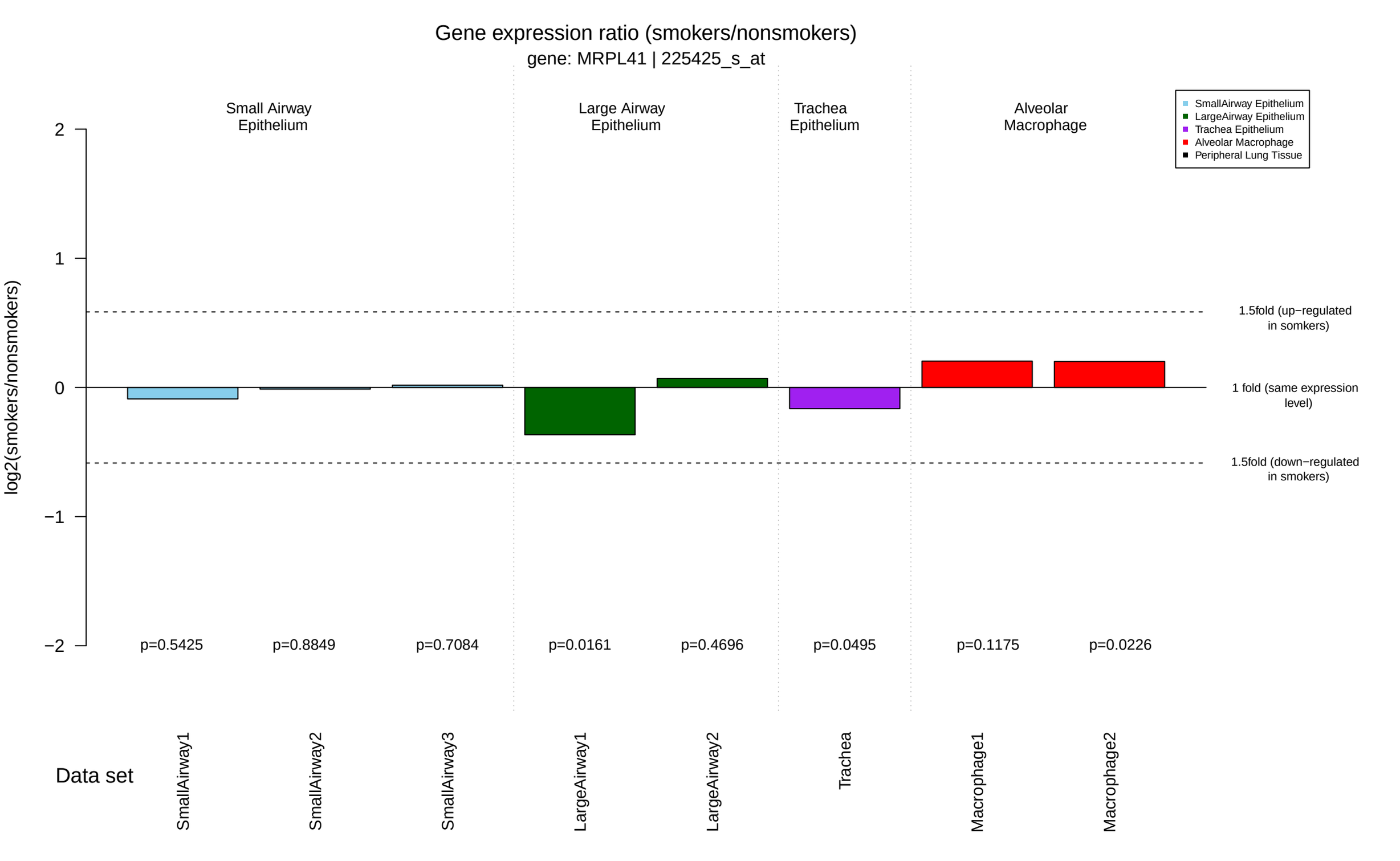
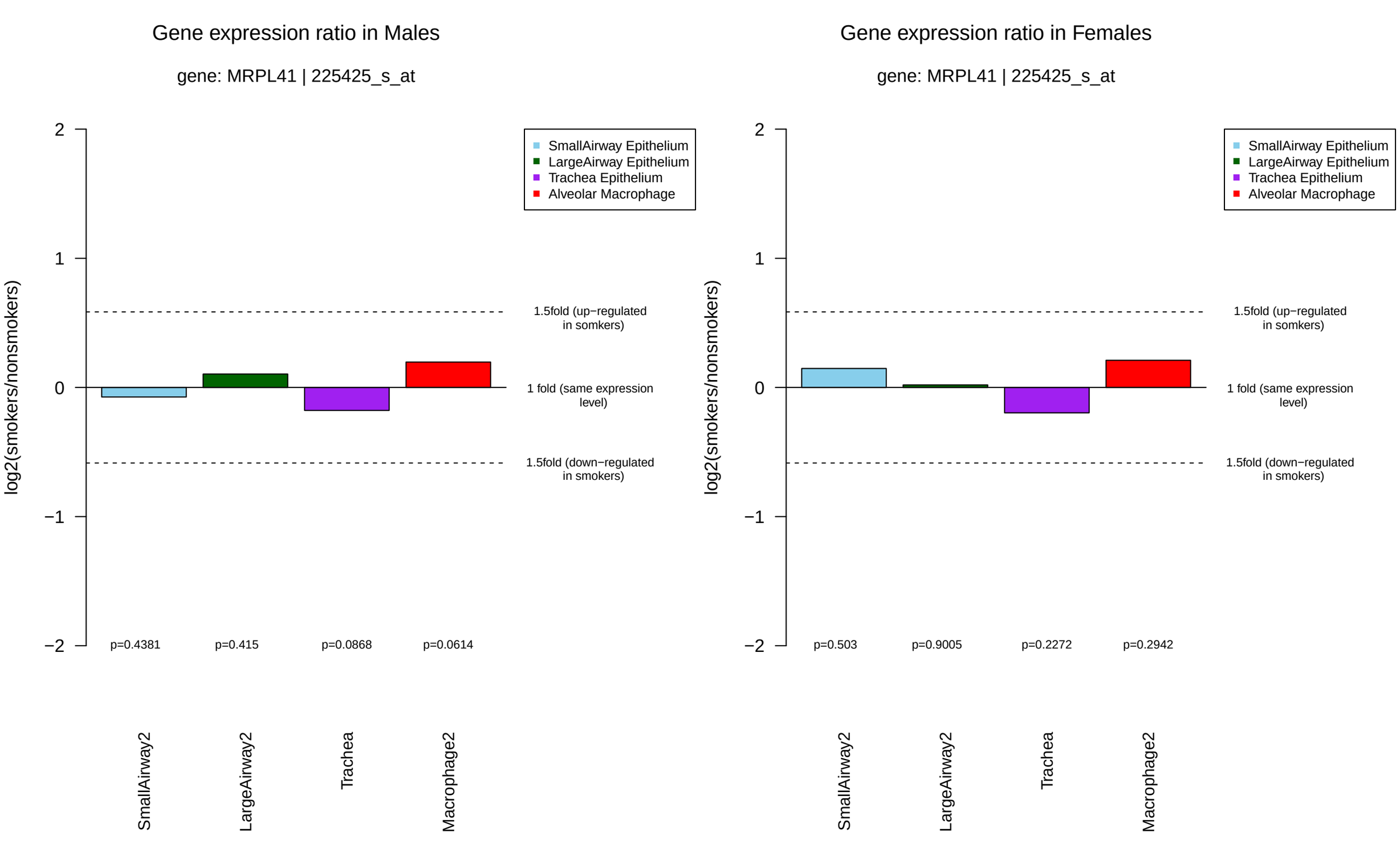
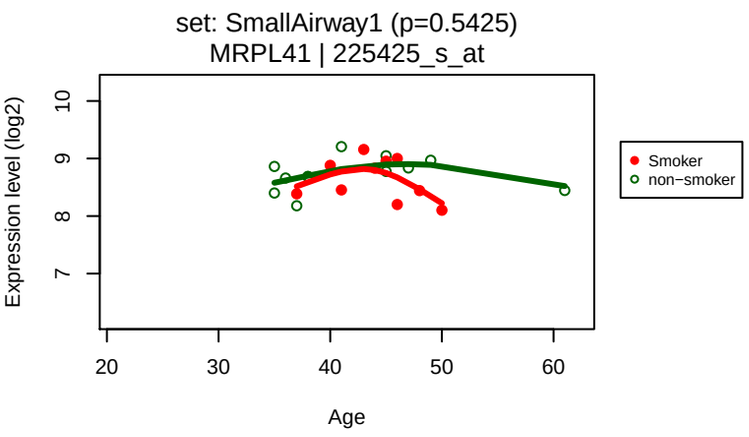
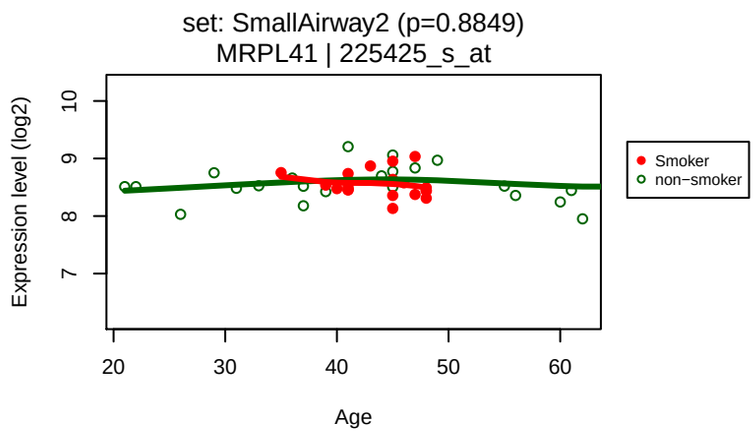
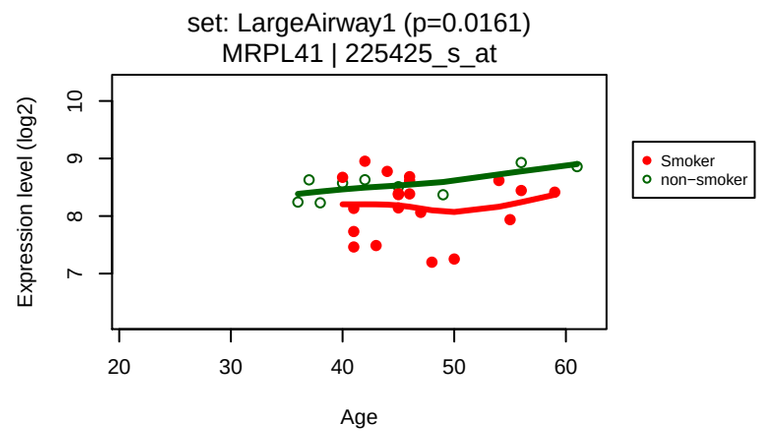

Gene expression of smokers and nonsmokers
gene: MRPL41 | 225425_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5425
p=0.8849
p=0.7084
p=0.0161
p=0.4696
p=0.0495
p=0.1175
p=0.0226
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

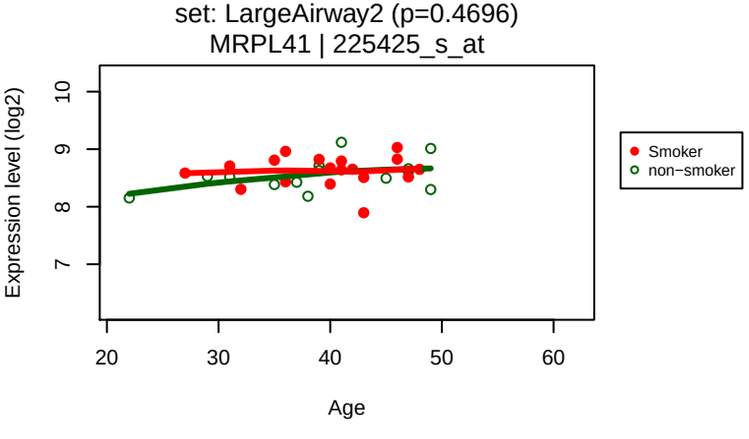

Gene expression ratio (smokers/nonsmokers)
gene: MRPL41 | 225425_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5425
p=0.8849
p=0.7084
p=0.0161
p=0.4696
p=0.0495
p=0.1175
p=0.0226
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

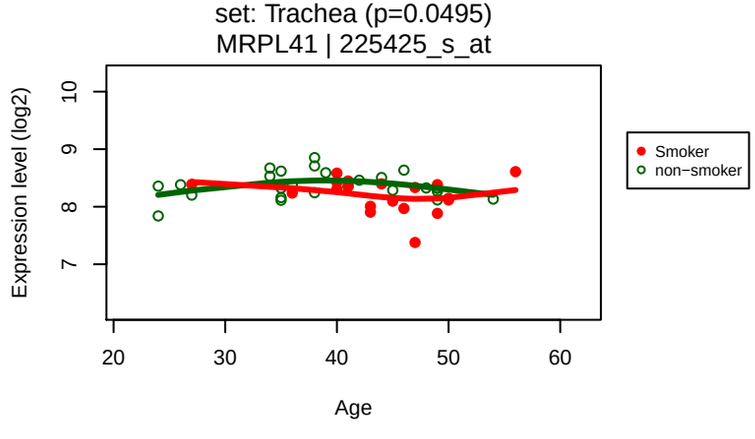

Gene expression ratio in Males
gene: MRPL41 | 225425_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4381
p=0.415
p=0.0868
p=0.0614
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL41 | 225425_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.503
p=0.9005
p=0.2272
p=0.2942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

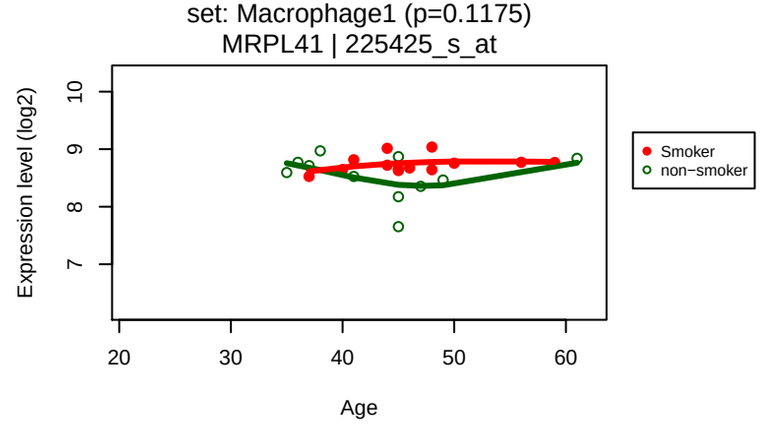

set: SmallAirway1 (p=0.5425)
MRPL41 | 225425_s_at
Expression level (log2)
Age
Smoker
non−smoker

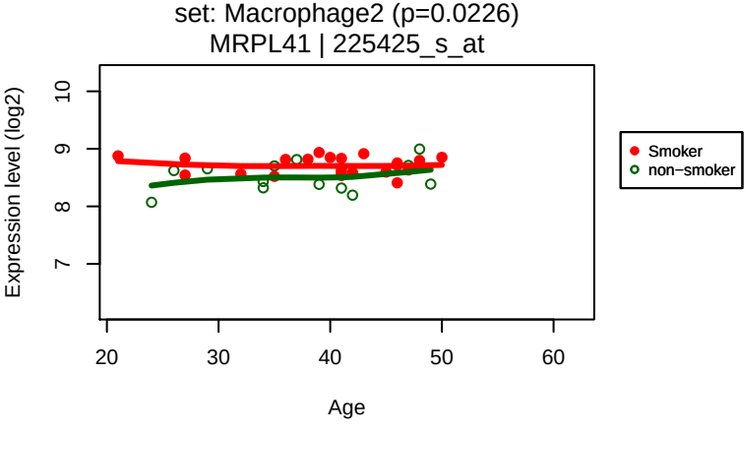

set: SmallAirway2 (p=0.8849)
MRPL41 | 225425_s_at
Expression level (log2)
Age
Smoker
non−smoker

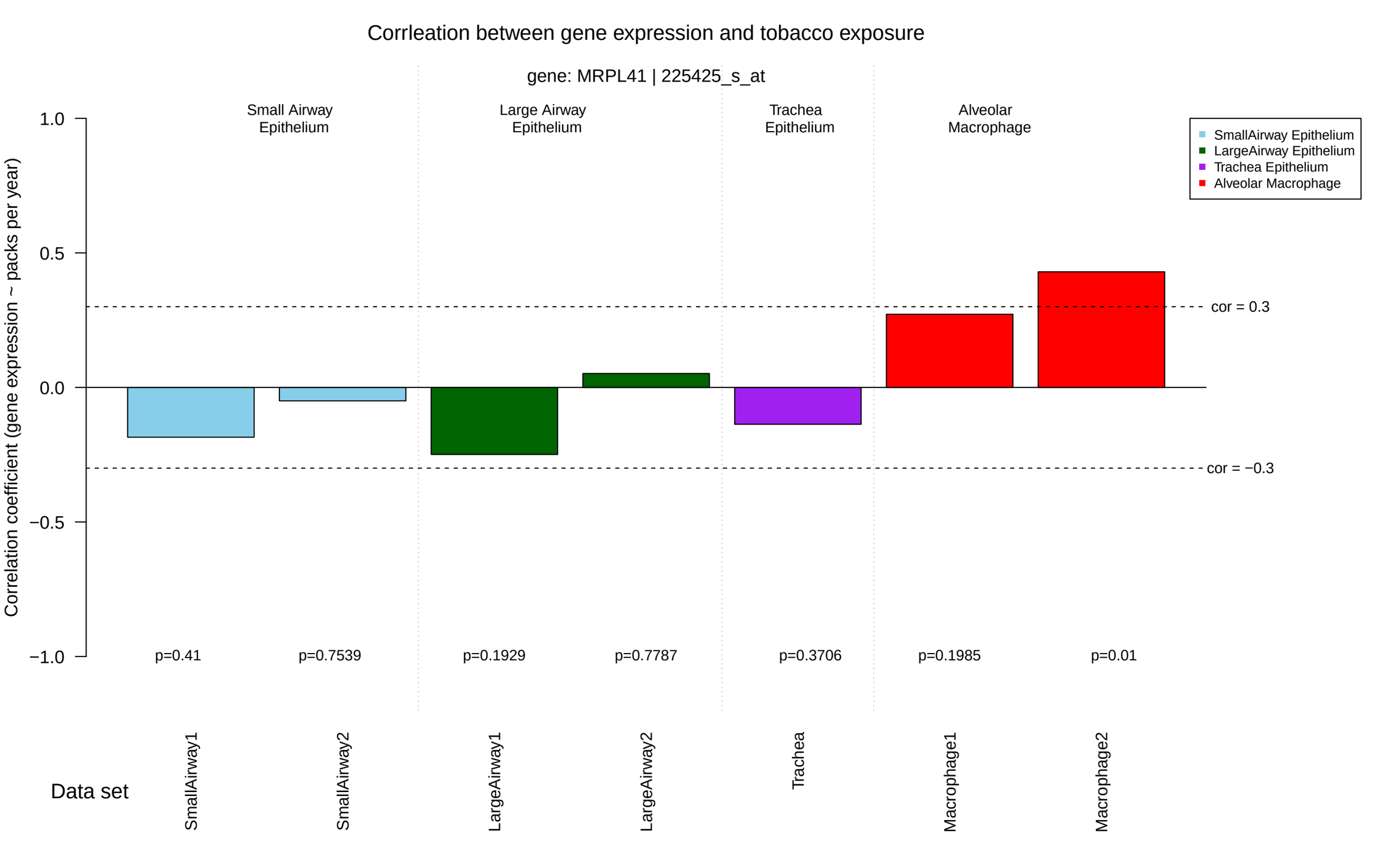

set: LargeAirway1 (p=0.0161)
MRPL41 | 225425_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4696)
MRPL41 | 225425_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0495)
MRPL41 | 225425_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1175)
MRPL41 | 225425_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0226)
MRPL41 | 225425_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL41 | 225425_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.41
p=0.7539
p=0.1929
p=0.7787
p=0.3706
p=0.1985
p=0.01
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2